

Concurrency II

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Based on slides by Troels Henriksen.

Inter-Thread Communication

Lets Thread a Program

- Fibonacci

- One thread per line

- Thread pools

- Condition Variables

Deadlock

Inter-Thread Communication

- So far we've only really talked about isolated processes and threads.
- Sharing data introduces a lot of problems with mutexes
- We can sidestep these problems with communication channels
- Most prominent in networking, but we can use them within a machine as well

Channels

- Any processes/thread can send a signal to any processes/thread that is listening for that signal
- This is necessary in processes as they cannot share data, this is the only way for them to do so.
- This is an alternative to shared data for threads.

Pipes

```
int pfd;  
pipe (pfd) ;
```

- A pipe is unidirectional with a dedicated read and write file descriptors.

Reading and Writing to a Pipe

```
write(pfd[1], char* buff, int size);  
read(pfd[0], char* buff, int size);
```

- We can read and write to these in a similar manner to any other file.
- Can't seek or move around the 'file', but we don't usually need to.

A Pipe Example

```
#define MSGLEN 16

int main(void) {
    char inbuf[MSGLEN];
    int p[2], i;
    if (pipe(p) < 0)
        return 1;
    if (fork() == 0) {
        write(p[1], "First", MSGLEN);
        write(p[1], "Second", MSGLEN);
    } else {
        for (i = 0; i < 2; i++) {
            read(p[0], inbuf, MSGLEN);
            printf("%s\n", inbuf);
        }
    }
}
```

Some Warnings

- This approach has a lot of overhead
- And is very easy to deadlock
- We will look at it again in more detail when its time for networking

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The Fibonacci Function

```
int fib (int n) {  
    if (n < 2) {  
        return 1;  
    } else {  
        return fib(n-1) + fib(n-2);  
    }  
}
```

- This is the slow formulation, but it will give us something to work with.
- Goal: apply fib() to all lines of a file

The `getline()` function

```
ssize_t getline(char **lineptr, size_t *n, FILE *stream);
```

- Allocates memory for us ('lineptr'), but it is our responsibility to 'free()' it when we are *completely* done.
- Returns the size of the line, and stores the size of the memory underlying it in 'n'.
- We must free the line when we are done with 'getline()'.

Using getline()

```
int main() {
    char *line = NULL;
    ssize_t line_len;
    size_t buf_len = 0;

    while ((line_len = getline(&line, &buf_len, stdin)) != -1)
        int num = atoi(line);
        printf("fib(%d) = %d\n", num, fib(num));
    }

    free(line);
}
```

The `atoi()` function

```
int atoi(const char *nptr);
```

- Returns integer represented by a string.

Is it fast?

```
time ./fibs < fibs-huge.input  
fib(40) = 165580141  
fib(41) = 267914296  
fib(42) = 433494437  
fib(43) = 701408733  
fib(45) = 1836311903
```

```
real    0m12.314s  
user    0m12.306s  
sys     0m0.006s
```

- Depends

Could it be faster?

- Yes - this program uses only a single thread, and my machine has fourteen cores.

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The thread function

```
void* fib_thread(void* arg) {  
    char *line = arg;  
    int n = atoi(line);  
    printf("fib(%d) = %d\n", n, fib(n));  
    free(arg);  
    return NULL;  
}
```

Changes to `main()`

```
int i = 0;
pthread_t threads[200000]; // arbitrary

while ((line_len = getline(&line, &buf_len, stdin)) != -1) {
    pthread_create(&threads[i], NULL, fib_thread, strdup(line))
    i++;
}

for (int j = 0; j < i; j++) {
    pthread_join(threads[j], NULL);
}
```

- Note the `strdup()` - this copies the line to avoid a race condition.

Is it faster?

```
time ./fibs > /dev/null < fibs-huge.input
```

```
real    0m12.314s
user    0m12.306s
sys     0m0.006s
```

```
time ./fibs-mt > /dev/null < fibs-huge.input
```

```
real    0m6.625s
user    0m12.250s
sys     0m0.006s
```

- We get a notable improvement

But...

```
time ./fibs > /dev/null < fibs-verytiny.input
```

```
real    0m0.012s
```

```
user    0m0.007s
```

```
sys     0m0.006s
```

```
time ./fibs-mt > /dev/null < fibs-verytiny.input
```

```
real    0m0.262s
```

```
user    0m0.053s
```

```
sys     0m0.388s
```

- Spawning a thread is `expensive` (relatively).

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Amortising thread startup cost

- It is often too slow to start a new thread for every piece of work.
- For compute-bound work, we only need one thread per CPU core.

Solution: Thread pools

- A *thread pool* is a collection of *worker threads* that wait for tasks.
- When a task is submitted, a thread is awoken, performs the task, then goes back to waiting for more.

Complex topic

- How big is the pool?
- How flexible?
- Do we use thread affinity?
- *We will only lightly touch on these concerns in the following.*

Creating threads for the pool is easy

```
// The number of processors.  
int num_threads = sysconf(_SC_NPROCESSORS_ONLN);  
  
// Make space for that many threads.  
pthread_t *threads = malloc(num_threads*sizeof(pthread_t));  
  
// Then launch them.  
for (int i = 0; i < num_threads; i++) {  
    pthread_create(&threads[i], NULL, worker, NULL);  
}
```

But how do we submit work?

- Pipes would not work here, because multiple threads could read from the same pipe.
- A line of input is bigger than one byte.

Global shared variables

```
// If not NULL, a line is ready to be processed.
```

```
char *volatile line = NULL;
```

```
// Lock before accessing 'line'.
```

```
pthread_mutex_t line_mutex = PTHREAD_MUTEX_INITIALIZER;
```

```
// If 1, threads should shut down.
```

```
volatile int die = 0;
```

The thread function

```
void* worker(void* arg) {
    arg=arg;
    int done = 0;
    while (!done) {
        char *my_line = NULL;
        assert(pthread_mutex_lock(&line_mutex) == 0);
        if (line == NULL && die) {
            done = 1;
        }
        if (line != NULL) {
            my_line = line;
            line = NULL;
        }
        assert(pthread_mutex_unlock(&line_mutex) == 0);
        if (my_line != NULL) {
            int n = atoi(my_line);
            printf("fib(%d) = %d\n", n, fib(n));
            free(my_line);
        }
    }
    return NULL;
}
```

The line-reading loop

```
while ((line_len = getline(&my_line, &buf_len, stdin)) != -1)
    int done = 0;
    while (!done) {
        assert(pthread_mutex_lock(&line_mutex) == 0);
        if (line == NULL) {
            line = strdup(my_line);
            done = 1;
        }
        assert(pthread_mutex_unlock(&line_mutex) == 0);
    }
}

die = 1;
```

Synchronisation by busy-waiting

- Worker threads spin in a lock/unlock-loop waiting for `line` to be non-NULL.
- The `main()` function spins in a lock/unlock-loop waiting for `line` to be NULL.

This is wasteful!

- When a worker thread sets `line` to NULL, it should *signal* the main thread that it can now store a new line.
- Similarly, the main thread should signal the worker threads when a line becomes available.
- This is where we use condition variables

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Initialisation

```
pthread_cond_t line_cond = PTHREAD_COND_INITIALIZER;
```

Signaling

```
int pthread_cond_signal(pthread_cond_t *cond);  
int pthread_cond_broadcast(pthread_cond_t *cond);
```

Waiting

```
int pthread_cond_wait(pthread_cond_t *cond,  
                        pthread_mutex_t *mutex);
```

- Blocks until another thread calls `pthread_cond_signal()`.
- The `mutex` *must* be locked when we call `pthread_cond_wait()`.
- Will be unlocked while the thread sleeps, and locked again when `pthread_cond_wait()` returns.
- *Spurious wakeups* may occur. ("MESA semantics")

Using condition variables in the worker threads

```
void* fib_thread(void* arg) {
    arg=arg;
    int done = 0;

    while (!done) {
        char *my_line = NULL;
        assert(pthread_mutex_lock(&line_mutex) == 0);

        if (line == NULL && !die) {
            pthread_cond_wait(&line_cond, &line_mutex);
        } else if (line == NULL && die) {
            done = 1;
        } else if (line != NULL) {
            my_line = line;
            line = NULL;
            pthread_cond_broadcast(&line_cond);
        }

        assert(pthread_mutex_unlock(&line_mutex) == 0);

        if (my_line != NULL) {
            int n = atoi(my_line);
            printf("fib(%d) = %d\n", n, fib(n));
            free(my_line);
        }
    }

    return NULL;
}
```

And in the main thread

```
while ((line_len = getline(&my_line, &buf_len, stdin)) != -1)
    int done = 0;
    while (!done) {
        assert(pthread_mutex_lock(&line_mutex) == 0);
        if (line == NULL) {
            line = strdup(my_line);
            pthread_cond_signal(&line_cond);
            done = 1;
        } else {
            pthread_cond_wait(&line_cond, &line_mutex);
        }
        assert(pthread_mutex_unlock(&line_mutex) == 0);
    }
}
```

- We still have the while-loop, but now it likely runs for much fewer iterations.

Another alternative: futures

- A *future* (sometimes *promise*) is a value that is being computed asynchronously. We may ask for the *value* of the future, which will block until it is ready.
- Not supported directly by POSIX threads.
- but `pthread_join()` is almost this model if you squint a bit.

Pseudocode for Fibonacci with futures:

```
def fib(n):  
    if n < 2:  
        return 1  
    x = future fib(n-1)  
    y = future fib(n-2)  
    return x.get() + y.get()
```

Why futures?

- A future may be evaluated in parallel, thus speeding up our program.
- They may also do other blocking non-CPU tasks, like network requests.
- *Most importantly:* Futures, if used correctly, are *deterministic*.
- (And they are not that hard to use correctly.)
- Futures are probably the simplest way to get a bit of parallelism or concurrency in your programs.

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Deadlock Avoidance

- Deadlock can occur any time we have a blocking operation
 - ▶ Network communications
 - ▶ Mutexes
- But if it *can* occur, we *must* assume it *will*
- Lets consider a deadlock in a simple counting program...

Deadlocking example: `main()`

```
pthread_mutex_t mutex[2] = {
    PTHREAD_MUTEX_INITIALIZER,
    PTHREAD_MUTEX_INITIALIZER
};
int cnt = 0;

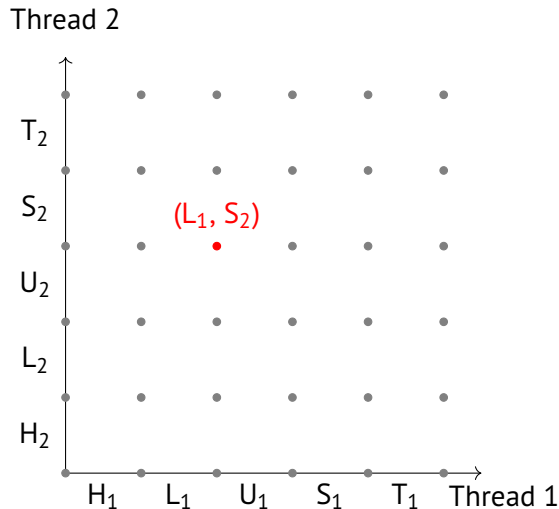
int main()
{
    pthread_t tid[2];
    pthread_create(&tid[0], NULL, count, (void*) 0);
    pthread_create(&tid[1], NULL, count, (void*) 1);
    pthread_join(tid[0], NULL);
    pthread_join(tid[1], NULL);
    printf("cnt=%d\n", cnt);
    exit(0);
}
```

Deadlocking example: count ()

```
void *count(void *vargp)
{
    int id = (int) vargp;
    for (int i = 0; i < 1000; i++) {
        pthread_mutex_lock(&mutex[id]);
        pthread_mutex_lock(&mutex[1-id]);
        cnt++;
        pthread_mutex_unlock(&mutex[id]);
        pthread_mutex_unlock(&mutex[1-id]);
    }
    return NULL;
}
```

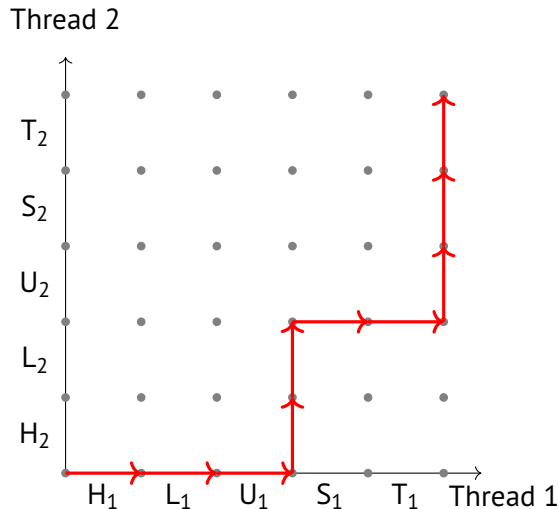
- Here we are claiming two locks at the same time
- And each thread claims them in an opposing order

Process Graphs



- A *progress graph* depicts the discrete *execution state space* of concurrent threads.
- Each axis corresponds to the sequential order of instructions in a thread.
- Each point corresponds to a possible *execution state* (Inst1, Inst2).
- E.g., (L_1, S_2) denotes state where thread 1 has completed L_1 and thread 2 has completed S_2 .

Trajectories in Process Graphs

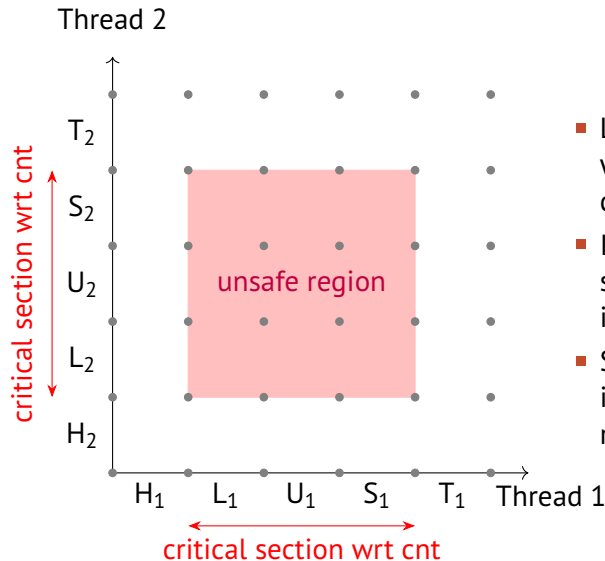


- A *trajectory* is a sequence of legal state transitions that describes one possible concurrent execution of the threads.
- Example: $H_1, L_1, U_1, H_2, L_2, S_1, T_1, U_2, S_2, T_2$

Enforcing Mutual Exclusion

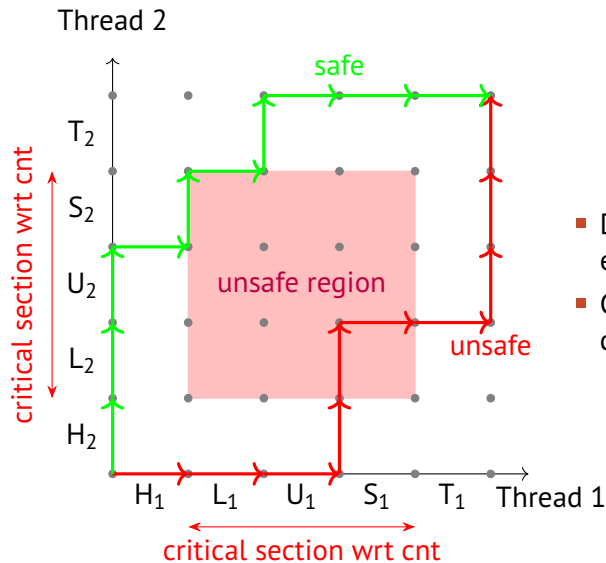
- Question: How can we guarantee a safe trajectory?
- Answer: We must synchronize the execution of the threads so that they can never have an unsafe trajectory.
 - ▶ i.e., need to guarantee mutually exclusive access for each critical section.
- Classic solution:
 - ▶ Semaphores (Edsger Dijkstra)
- Other approaches
 - ▶ Mutexes and condition variables from Pthreads
 - ▶ Monitors (Java) (boring languages are outside our scope)

Critical Sections and Unsafe Regions



- L, U, and S form a critical section with respect to the shared variable cnt
- Instructions in critical sections (wrt. some shared variable) should not be interleaved
- Sets of states where such interleaving occurs form unsafe regions

Critical Sections and Unsafe Regions



- Def: A trajectory is *safe* iff it does not enter any unsafe region
- Claim: A trajectory is correct (wrt cnt) iff it is safe

Deadlocking example: count ()

```
void *count(void *vargp)
{
    int id = (int) vargp;
    for (int i = 0; i < 1000; i++) {
        pthread_mutex_lock(&mutex[id]);
        pthread_mutex_lock(&mutex[1-id]);
        cnt++;
        pthread_mutex_unlock(&mutex[id]);
        pthread_mutex_unlock(&mutex[1-id]);
    }
    return NULL;
}
```

Thread0

P(mutex₀)

P(mutex₁)

cnt++

V(mutex₀)

V(mutex₁)

Thread1

P(mutex₁)

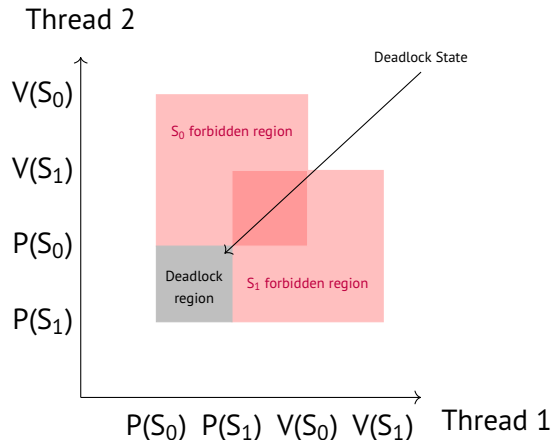
P(mutex₀)

cnt++

V(mutex₁)

V(mutex₀)

Deadlock Visualised in a Process Graph



- Locking introduces the potential for deadlock: waiting for a condition that will never be true
- Any trajectory that enters the deadlock region will eventually reach the deadlock state, waiting for either s_0 or s_1 to become nonzero
- Other trajectories luck out and skirt the deadlock region
- Unfortunate fact: deadlock is often nondeterministic (race)

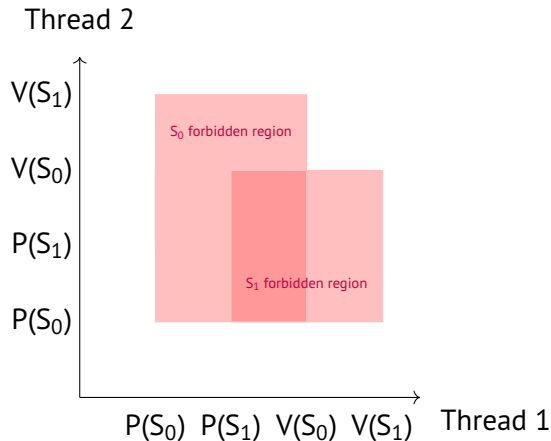
Avoiding Deadlock

```
void *count(void *vargp)
{
    int id = (int) vargp;
    for (int i = 0; i < 1000; i++) {
        pthread_mutex_lock(&mutex[0]);
        pthread_mutex_lock(&mutex[1]);
        cnt++;
        pthread_mutex_unlock(&mutex[0]);
        pthread_mutex_unlock(&mutex[1]);
    }
    return NULL;
}
```

<i>Thread0</i>	<i>Thread1</i>
P(mutex ₀)	P(mutex ₀)
P(mutex ₁)	P(mutex ₁)
cnt++	cnt++
V(mutex ₀)	V(mutex ₀)
V(mutex ₁)	V(mutex ₁)

- Note we've changed the ordering of claiming the mutex
- Now its the same for both

Avoiding Deadlock in a Process Graph



- No way for trajectory to get stuck
- Processes acquire locks in same order
- Order in which locks released immaterial
 - ▶ Why?

Drawing a process graph

- Easy to draw! (mostly).
 - ▶ Most applications are 'symmetrical', and if they aren't they probably should be.
- We can ignore any non-blocking code.
 - ▶ Anything that completes in a finite time can be ignored.
- We only need to plot two axis' (mostly).
 - ▶ It doesn't matter if we have 2 processes or 2 million. The same logical dependency exists between them.
- When in doubt, draw a diagram!
 - ▶ We often can't prove that we have avoided deadlock, but a diagram can be a short-hand for showing how its impossible, if our diagram reflect our code.